

Nifty 100 Financial Intelligence Platform

Analyst User Guide & API Reference Manual

1. System Overview & Architecture

The Nifty 100 Financial Intelligence Platform is an end-to-end institutional-grade analytics engine combining automated data pipelines, rule-based NLP extraction, machine learning clustering, a high-performance FastAPI REST backend, and an interactive Streamlit web dashboard.

2. Live Cloud Deployments

- **Frontend Dashboard (Streamlit Cloud):** <https://n100financialintelligenceplatform-9kqrcgnirnyoseylf8ztw8.streamlit.app/>
- **Backend REST API (Render):** <https://n100-financial-intelligence-platform-api.onrender.com/docs>

3. Streamlit Dashboard Navigation

- **Home:** Macro-level market overview, index KPI summaries, and sector distribution charts.
- **Profile:** Deep dive into individual company balance sheets, ROCE/ROE trajectories, and P&L; histories.
- **Screener:** Custom multi-factor screening using real-time metric sliders and financial preset filters.
- **Peers:** Industry group peer benchmarking and interactive multi-axis radar comparison charts.
- **Trends:** Multi-metric overlay analysis for tracking YoY financial growth trajectories.
- **Clusters (New):** Machine learning KMeans financial archetypes (\$k=5\$), elbow curves, and correlation matrices.

4. Generating & Downloading PDF Tearsheets

Users can navigate to the **Reports** module inside the dashboard to instantly preview and download programmatically compiled 2-page company tearsheets and 11 sector intelligence summaries generated via ReportLab.

5. FastAPI REST API & Example cURL Commands

The backend server runs on FastAPI and exposes 16 documented endpoints. Interactive API documentation is available at <https://n100-financial-intelligence-platform-api.onrender.com/docs>.

Health Check:

```
curl -X GET "https://n100-financial-intelligence-platform-api.onrender.com/api/v1/health"
```

Fetch All Companies:

```
curl -X GET "https://n100-financial-intelligence-platform-api.onrender.com/api/v1/companies?sector=IT"
```

Download Company Tearsheet PDF:

```
curl -X GET  
"https://n100-financial-intelligence-platform-api.onrender.com/api/v1/companies/INFY/tearsheet"  
--output infy_tearsheet.pdf
```

6. Troubleshooting & Support

- **Port Conflicts:** Ensure local ports are free if testing backend changes locally.
- **Missing Data Errors:** Run the ETL ingestion script and clustering module (`python src/analytics/clustering.py`) if local artifacts are missing.